Kuku Promo Finder

AI Generalist Assignment · Problem 1: Find the Promos · a hybrid rule + (optional) LLM tool that reads microdrama screenplays and returns ranked 20-60 s promos with exact timestamps

1. Problem and why I picked it

An editor today watches every episode to pick promo moments, one show at a time, while new shows ship every week. I chose this problem because the input pack is real, the output is checkable (timestamps either line up or they don't), and the business pain is direct: promos are how new viewers arrive. The screenplays already carry a timestamp for every shot, line and action, so most of the search can be done exactly, in code, with AI judgment saved for the part that needs it.

2. Assumptions

• The screenplay timestamps match the final cut, so a promo is defined by script timestamps (no video was available). • A clip may cut only on a camera-shot boundary and never mid-line, which is how an editor cuts. • "First 3 seconds" = blocks starting in the first 3.0 s of the first segment; the "ending" = the last ~6 s. • Multi-segment promos join a short hook sting to a later stretch, in story order (so the sting can't spoil the ending), from the same or previous episode, sharing a character. • "Across all episodes" means coverage: up to 2 promos per episode, each tagged with a confidence tier. • Delivery tags such as (Furious) or (Cold) are a usable proxy for performance intensity.

3. Architecture

4. How it works

Parse each episode into typed blocks (CAMERA, ACTION, SOUND FX, dialogue with speaker and delivery tag), with seconds-precision times. **Generate** every valid start/end pair of 20-60 s (12,127 singles), plus 366 stitched candidates for strong endings with weak openings. **Score** each on five criteria with logged evidence (e.g. "+0.25 stakes in final line (gallows)"). **Select**: hook and cliffhanger gates (≥0.30), then greedy non-maximum suppression (≤15% shared screen time, no two promos with the same opening, ≤2 per episode). **Optional LLM**: the top 40 are re-judged by OpenAI, Gemini or a local Ollama model (provider set by one environment variable; stdlib only, cached, falls back to rules on any error) and blended 50/50. **Explain**: hook, cliffhanger and "why" are templates filled with the actual script lines. No text is invented.

5. Ranking methodology

Criterion	Weight	Main signals (all points logged as evidence)
Hook (first 3 s)	0.27	physical event (slap, grab, lunge) · shocking line (married, poison, gallows) · explosive delivery · exclamation or question; minus for opening on a reply or an establishing shot
Cliffhanger (last ~6 s)	0.27	question **whose answer exists in the script but is cut** · stakes or threat in the last line · trailing line · tension cue · arrival · cut on the episode's own ending; minus for resolution words, energy drop, dead air
Emotional intensity	0.16	mean and peak delivery intensity (3-level lexicon from the 620 tag words in the pack) + physical beats
Standalone	0.15	minus off-screen names and backstory phrases; plus relationship anchors (husband, sister, Duke)
Narrative clarity	0.15	dialogue covers 35-85% of the clip, 2-3 speakers, turn-taking; minus scene changes
Duration	gate	hard 20-60 s; ×1.0 inside 25-50 s, down to ×0.9 at the limits

Tiers: ≥0.75 **post-first**, ≥0.60 **review**, otherwise **weak**. Ties break on promo ID, so the ranking is deterministic.

6. Actual results on the 50-episode pack (rules only; no LLM was reachable)

50/50 episodes parsed (3,727 blocks; 0 warnings after a fix) · 12,493 candidates · 2,720 pass the gates · **95 promos selected** (18 post-first, 61 review, 16 weak), 14 multi-segment · durations 21.6-57.5 s (median 34.88) · **15/15 automated checks pass**, including two runs giving identical output · 7.6 s on one CPU core.

#	Episode · segments	Dur	Score	Hook	Cliffhanger
1	E36 02:43.00-03:20.75	38s	0.89	Mother grabs his coat: "You cannot go back there alone!"	"...their dead Duke is coming home." (episode end)
2	E15 00:07.50-00:11.80 + E15 00:49.50-01:26.00	41s	0.89	Juliana bursts in: "Stop! What are you doing?!"	"You want to fight us, veteran?" (cut before the fight)
3	E28 01:10.50-01:15.30 + E28 01:47.80-02:18.50	36s	0.88	Catherine yanks the locket off Juliana's neck	"...before he sends you to the gallows."
4	E32 01:59.50-02:34.38	35s	0.85	Catherine raises a branding iron: "Silence!"	"...let's see how brave you are when it touches your skin!"
5	E29 00:56.60-01:36.10	40s	0.83	"You are doing this because I saved the Duchess!"	Cell door slams; alone: "Mother..." (softer ending)
6	E16 00:14.30-00:44.20	30s	0.81	A thug lunges at Alexander with a weapon	"Who sent you?" (answer withheld)

Post first: #1-#4 are the only promos strong at **both** ends (physical open, stakes spoken at the cut). #5-#8 open just as hard but end softer. Full list: outputs/ranked_promos.csv.

7. Quality evaluation, failure cases and limitations

Evidence	Result
Automated validation	15/15 checks pass: timestamps inside episodes, cuts on shot boundaries and never mid-line (all 106 segments also re-checked against the raw text), 20-60 s, no shared openings, ≤15% overlap, reproducible. **Ablation:** with the writers' "cliffhanger" SFX notes (in 22 scripts) ignored, the top 10 is unchanged and 88 promos are shared.
AI-assisted review	Claude read the top 20 against the script: 9 post as-is, 8 usable after a trim, 3 not post-first-worthy.
Human / editor review	**Not done yet**; no human labels or accuracy figures exist. outputs/review_sheet.csv is ready for it.

Failure / limitation	Real example from this run
Keyword out of context	#17 E38 gets a "shocking line" bonus for "no one should **die** hungry", a guard's kindness; it also opens on a reply ("Why?").
Standalone ignores pronouns	#15 E30: "we give them a reason to condemn **her**"; "her" is never seen, yet it scores 0.85.
End point a beat early	#8 E11 ends 3 s before the stronger line ("Going to strike a gentleman...?").
Weak-episode filler	16 weak-tier promos; episodes 14, 17, 19, 49 and 50 have nothing above weak (e.g. E19 exposition).
Hook saturation	many "Stop!" or scream openings score 1.0, so the top is separated mainly by the cliffhanger.
Script, not footage	performance, music and pacing are invisible; series-level spoilers are not modelled.

Fixed in the final audit (old numbers reproduced first): overlap was divided by segment instead of clip (a 3 s sting once removed Episode 17's only promo); one sting opened three top-20 promos; E24 began 1.3 s *after* its riding-crop strike; a stitch relied on a character who was only mentioned. **Human in the loop:** mark review_sheet.csv, list IDs to reject or pin in an overrides file, rerun with --overrides.

8. Scaling to every show

Rules cost about 0.14 s per episode on one core, so 1,000 episodes take about 2-3 minutes; shows run in parallel. The LLM sees only the shortlist: a measured ~570 input tokens per clip × 40 clips ≈ 25k tokens per show, cached, and free on local Ollama. The tool is format-driven, not show-driven: character names are learned per run, and a new show only needs --input. The editor's job shifts from watching every episode to approving about 20 post-first promos per show. Next steps: fit the weights to the editor's verdicts, add pronoun checks, try ±1-2 shots around each cut, allow one post-first promo per episode, and run the LLM judge on the shortlist.

9. AI tools used, and one place they got it wrong

Claude (Anthropic) helped design, write and test the code, read the script to check the top picks, and draft this document; I reviewed the decisions and ran the tool. **Where it went wrong:** its first keyword matcher used prefix matching, so "beg" fired on "**beg**gar's pocket", "explo" on "**explo**sive fury", "before" in "scrub floors before these commoners" counted as a deadline, and a riding crop that was only *held* scored as a slap. These inflated hook scores near the top of the list. Reading the top picks against the script exposed it; the fix was whole-word matching, events instead of props, and a regression test. It also missed that Episode 6 writes SCENE START: in capitals, until the quality checker flagged untimed scenes.